

Vehicle-Side Lightweight Conditional Anonymous Double Authentication Framework With Password Robustness in VANETs

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Abstract—The emergence of Vehicular Ad Hoc Networks (VANETs) has revolutionized modern transportation by enabling real-time and cooperative communication between vehicles and infrastructure. However, vehicular interactions operate across two layers: human–vehicle access and vehicle–network communication. This dual-layered nature introduces complex security and privacy challenges. Existing Conditional Privacy-Preserving Authentication (CPPA) schemes mainly focus on message-level anonymity and integrity. However, they often overlook systemic vulnerabilities across the entire trust lifecycle. These limitations include the inability to verify driver legitimacy during vehicle access, high computational overhead due to heavyweight cryptographic primitives, the lack of password update and recovery mechanisms, and insufficient resistance to physical key extraction attacks. To address these issues, we propose VLCADA, a Vehicle-side Lightweight Conditional Anonymous Double Authentication framework. VLCADA establishes a vertically integrated security architecture for intelligent vehicles. It unifies multi-factor driver authentication with efficient and conditionally anonymous message authentication. At the access layer, VLCADA combines identity–password credentials, biometric verification, and Physical Unclonable Functions (PUF). This ensures that only legitimate users can access and control the vehicle. At the communication layer, we design a Signer-side

Lightweight Conditional Anonymous Signature (SLCAS) scheme. This scheme shifts computationally intensive operations to an off-line phase, leaving only lightweight hash and XOR operations during real-time execution. In addition, we propose a Dynamic Password Mechanism (DPM) based on threshold secret sharing. This mechanism enables secure password update and recovery without trusted authority intervention. Furthermore, by integrating PUF with fuzzy extractors, VLCADA effectively mitigates noise and provides resistance against physical attacks. Comprehensive security analysis shows that VLCADA satisfies essential security requirements. Experimental results demonstrate that VLCADA reduces total computational overhead by up to 48.55% and communication cost by up to 32.61% compared with state-of-the-art schemes.

Index Terms—VANETs, authentication, conditional anonymous, password robustness, physical unclonable function.

I. INTRODUCTION

WITH the dedicated allocation of wireless spectrum and the evolution of standards such as Dedicated Short-Range Communications (DSRC), VANETs have become a foundational infrastructure for Intelligent Transportation Systems (ITS) and real-time vehicular communication environments [1]. VANETs enable vehicles to disseminate traffic information, facilitate safety-critical alerts, and support high-level applications such as cooperative autonomous driving and real-time navigation optimization [2]. For instance, autonomous vehicles rely on continuous synchronization with RoadSide Units (RSUs) to ensure seamless transitions during handovers between communication zones [3]. These communications are facilitated through both Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) channels, underpinned by an increasingly dense RSU infrastructure that forms a resilient network topology [4].

Despite these advancements, the sensitivity of vehicular data exchanged over V2V and V2I links exposes VANETs to a wide spectrum of security threats, including eavesdropping, message tampering, identity spoofing, and trajectory tracking [5]. Addressing these challenges requires robust authentication and data integrity mechanisms to ensure that only authorized vehicles can broadcast legitimate and verifiable messages [6]. While conventional solutions primarily target active attacks through public key cryptography and digital signatures, passive attacks, such as long-term vehicle tracking via static identifiers, continue to erode user privacy [7]. Although pseudonym-based schemes

Received 31 October 2025; revised 7 June 2026; accepted 8 June 2026. Date of publication 11 June 2026; date of current version 1 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62302070 and Grant 62072133, in part by the Zhejiang Provincial Natural Science Foundation of China under Grant ZCLZ25F0201, in part by Wenzhou Major Scientific and Technological Innovation Projects under Grant ZG2023028. (Corresponding authors: Hong Sun; Yining Liu.)

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Digital Object Identifier 10.1109/TDSC.2026.3702678

offer some protection [8], adversaries can still inject forged messages or link pseudonyms over time, highlighting the need for authentication frameworks that provide both conditional anonymity and misbehavior traceability. CPPA has emerged as a promising paradigm to reconcile these goals, enabling a Trusted Authority (TA) to revoke anonymity in cases of malicious behavior [9].

A. Motivation and Contributions

The rapid convergence of vehicular automation and intelligent connectivity is fundamentally transforming modern automotive systems. Contemporary vehicles are no longer isolated mechanical machines but integral components of a multi-layered cyber-physical ecosystem. This ecosystem comprises two inter-dependent interfaces:

- *Driver-Vehicle Interface*: Access to in-vehicle functionalities is mediated through digital credentials such as passwords, biometrics, or smartphone-based keys.
- *Vehicle-Network Interface*: Vehicles engage in secure V2V and V2I communications via VANETs, enabling cooperative driving and traffic coordination.

At the access control layer, traditional mechanical keys have been largely replaced by digital authentication mechanisms. However, most existing CPPA schemes [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28] are designed solely to secure communications, under the implicit assumption that any authenticated vehicle is operated by a legitimate driver. This assumption creates a critical vulnerability: if a user credential (e.g., password or token) is compromised, an attacker may gain unauthorized physical access to the vehicle and exploit its valid credentials to launch broader attacks. Even those CPPA protocols that integrate multi-factor authentication [29], [30], [31] (e.g., combining password and biometric verification) often overlook essential lifecycle management features such as password update, or recovery. Once credentials are leaked or lost, these schemes offer no secure remediation pathway, leaving the vehicle persistently exposed.

At the communication layer, vehicles must perform timely and secure data exchanges despite their inherent resource constraints in computation, memory, and energy. While many schemes adopt pairing-based operations [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], these techniques impose significant computational burdens that undermine real-time performance in VANETs. Furthermore, although these designs [10], [11], [12], [13], [14], [15], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29] often assume that secret keys are securely stored in Tamper-Resistant Devices (TRDs), practical adversaries have demonstrated the feasibility of physical attacks such as side-channel analysis, fault injection, and chip extraction to retrieve sensitive credentials. Such breaches can result in cloning or Sybil attacks, with malicious On-Board Units (OBUs) masquerading as legitimate vehicles. These challenges reveal a systemic flaw: authentication is treated as a set of isolated functions, rather than as a holistic, end-to-end secure lifecycle. Specifically:

- They [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28] fail to verify the legitimacy of the human driver at the point of entry.
- They [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31] require vehicles to perform pairing-based operations, which impose considerable computational overhead and lead to increased communication latency, thereby limiting their applicability in the latency-sensitive environment of VANETs.
- They [10], [11], [12], [13], [14], [15], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29] offer no protection against physical key extraction.
- They [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31] lack mechanisms for password update, recovery, or revocation.

Consequently, the security of the entire vehicle is weakened by assumptions of trust at either end of the authentication pipeline. Without a unified framework, adversaries can exploit this gap, either by gaining physical access to the vehicle and impersonating a legitimate source, or by compromising the OBU's credentials and transmitting forged messages.

To address these challenges, we propose **VLCADA**, which is the first architecture to unify both ends of the vehicular trust chain, integrating physical driver verification with conditional anonymous message authentication. It establishes a continuous trust pathway from physical access to wireless communication. Our main contributions are as follows.

- *Double-Authentication Architecture*: We propose a vertically integrated model that links driver-side access control with in-vehicle message authentication. Only verified drivers can operate the vehicle and initiate communications. This contrasts with existing CPPA schemes [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], which treat these domains separately.
- *Multi-Factor Security with Password Robustness*: VLCADA employs multi-factor authentication combining password, biometrics, and PUF. It is secure against up to $n-1$ compromised factors out of n . A dynamic password mechanism allows password update and recovery, addressing the limitation of prior works [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31].
- *Lightweight Vehicle-Side Design and Physical Security*: To accommodate OBU constraints, we offload computationally intensive operations to an offline preparation phase. Runtime operations rely only on lightweight primitives (e.g., hash, XOR). Additionally, we integrate PUF and fuzzy extractor to defend against physical tampering, improving upon prior CPPA schemes [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31].
- *Formal Security Analysis and Performance Evaluation*: Extensive security analyses and performance evaluations

TABLE I
COMPARISON OF AUTHENTICATION SCHEMES

Schemes	Acc-Aut	V2R-Aut	Veh-Lig	Con-Ano	Mul-Fac-Sec	Pas-Rec	Pas-Upd	Res-Phy	Res-Gue	Res-Rep
PKI-based methods [10]–[13]	○	●	○	●	○	○	○	○	○	●
ID-based method [29]	●	●	○	●	○	○	○	○	○	●
ID-based methods [14], [15], [17], [18]	○	●	○	●	○	○	○	○	○	●
ID-based method [30]	●	●	○	●	○	○	○	●	○	●
ID-based method [16]	○	●	○	●	○	○	○	●	○	●
ID-based method [31]	●	●	●	●	○	○	●	○	○	●
CLC-based methods [19], [20], [23], [25]–[28]	○	●	○	●	○	○	○	○	○	●
CLC-based methods [21], [24]	○	●	○	●	○	○	○	○	○	○
CLC-based method [22]	○	●	○	○	○	○	○	○	○	○
Ours	●	●	●	●	●	●	●	●	●	●

Acc-Aut: Access authentication, V2R-Aut: V2R authentication, Veh-Lig: Vehicle-side lightweight, Con-Ano: Conditional anonymity, Mul-Fac-Sec: Multi-factor security, Pas-Rec: Password recover, Pas-Upd: Password update, Res-Phy: Resistance to physical attacks, Res-Gue: Resistance to guess attacks, Res-Rep: Resistance to replay attacks.
● implemented; ○ not implemented or considered.

demonstrate that VLCADA not only meets the stringent security requirements but also minimizes computational and communication overhead compared to current approaches [18], [25], [26], [27], [28], [29], [31], [32], [33], [34], [35], [36].

In sum, VLCADA is the first solution to consolidate human-to-vehicle access control, efficient vehicular message authentication, credential robustness, and tamper resistance into a single, cohesive framework. It represents a paradigm shift from isolated CPPA components to a complete, vertically integrated architecture for future-ready vehicular security.

B. Paper Structure Overview

Section II reviews relevant literature; Section III outlines the fundamental concepts; Section IV presents the system model; Section V details the design and development of VLCADA; Section VI analyzes the security properties; Section VII evaluates performance metrics; and Section VIII concludes with a summary of our work.

II. RELATED WORK

The exponential growth of VANETs has revolutionized modern transportation while simultaneously presenting significant security and privacy challenges. To overcome these issues, numerous authentication methods [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31] have been developed. In this section, we review these methods by classifying them into three principal categories: Public Key Infrastructure (PKI)-based [10], [11], [12], [13], Identity-based (ID-based) [14], [15], [16], [17], [18], [29], [30], [31], and CertificateLess Cryptography (CLC)-based schemes [19], [20], [21], [22], [23], [24], [25], [26], [27], [28]. Table I presents a comparative analysis that highlights the strengths and weaknesses.

A. PKI-Based Authentication Schemes

PKI has traditionally underpinned secure communications in VANETs by using digital certificates for entity authentication and message integrity. Hubaux et al. introduced the inaugural CPPA protocol for VANETs in 2003 [37], and this was followed by Raya et al.'s PKI-based CPPA approach in 2007 [10]. These protocols utilize multiple public/private key pairs and pre-installed certificates in vehicles' OBUs to mask real identities. Although widely implemented, they encounter problems such as substantial computational overhead, intricate certificate management, and scalability constraints.

In 2008, Lu et al. proposed an innovative CPPA method that employs anonymous certificates [11], where vehicles acquire short-lived certificates from RSUs as they travel. To preserve privacy, vehicles must regularly secure new certificates; however, this frequent interaction with RSUs proves both inefficient and impractical. Addressing these shortcomings, Lin et al. introduced a blockchain-based authentication protocol in 2021 [12] that leverages the PKI framework to alleviate key escrow concerns and boost performance. Later, in 2023, Zhou et al. presented a blockchain-enhanced CPPA scheme for VANETs [13] that surpassed earlier techniques in signing speed, verification processes, and bandwidth consumption. Nonetheless, challenges such as high storage demands and the heavy burden of certificate revocation and management continue to impede the widespread adoption of PKI-based approaches in extensive VANETs.

B. ID-Based Authentication Schemes

In an effort to improve message authentication efficiency and diminish reliance on conventional public-key certificates, researchers have proposed several ID-based schemes [14], [15], [16], [17], [18], [29], [30], [31]. In these designs, a Key Generation Center (KGC) assigns each user a private key derived from their unique identifier, which simultaneously functions

as the public key [38]. This arrangement simplifies the correlation between public keys and identities, thereby facilitating more effective entity verification. Early implementations of ID-based methods [14], [15], [17] frequently depended on bilinear pairings, operations known for their computational intensity that can hinder network performance. To mitigate this, He et al. developed an ID-based protocol that forgoes bilinear pairings, enhancing operational efficiency while maintaining robust security against adaptive chosen-signature and chosen-message attacks under the random oracle model [29]. Later improvements [16], [18], [30], [31] further refined these schemes to function without pairing operations.

In 2021, Wei et al. proposed a new CPPA approach that utilizes message recovery alongside secret sharing techniques to update the Secret Sharing Keys (SSKs) maintained in the Hardware Security Modules (HSMs) of valid vehicles [30]. This scheme, however, was subsequently shown to be susceptible to forgery attacks, allowing adversaries to produce legitimate signatures with ease [39]. That same year, Zhang et al. presented a CPPA method based on the Chinese Remainder Theorem (CRT) [16]. This technique removed the requirement to preload primary keys in vehicles' Trusted Platform Modules (TPDs), instead relying on dynamic key updates distributed by TAs via CRT-based group key mechanisms.

In 2023, Xu et al. introduced a distributed authentication protocol that capitalizes on the collaboration between RSUs and cooperative vehicles [18]. Although this approach achieved minimal overhead in both authentication and communication, its reliance solely on computing power for selecting cooperative vehicles could permit high-powered yet untrustworthy vehicles to compromise the system. Most recently, in 2024, Zhang et al. developed an ID-based batch multisignature protocol that disseminates SSK update messages over public channels [31]. Upon receiving these messages, vehicles verify their authenticity before updating their SSKs accordingly. Despite these advancements, ID-based schemes still grapple with the inherent key escrow problem. Since a trusted KGC generates all private keys, a breach of the KGC could expose every user's private key, leading to extensive security risks. This critical vulnerability underscores the need for further research to develop more secure authentication frameworks.

C. CLC-Based Authentication Schemes

To overcome the limitations of PKI-based and ID-based methods, several CLC-based CPPA protocols [19], [20], [21], [22], [23], [24], [25], [26], [27], [28] have been proposed for VANETs. These protocols eliminate the requirement for digital certificates and lessen dependence on a single trusted authority by combining a partial private key with a user-selected secret value. As a result, even if an adversary compromises one component, the complete secret key remains secure [40]. Horng et al. introduced a CLAS based on bilinear pairing; however, it is prone to malicious-but-passive KGC attacks [19]. In response, several pairing-based constructions [20], [21], [22] have been proposed, although they often incur inefficiencies due to the computational overhead of pairing operations. To enhance both communication and computational efficiency, researchers have subsequently

developed multiple pairing-free CLAS protocols [23], [24], [25], [26], [27], [28].

In 2018, Cui et al. presented an innovative authentication protocol [23]. Despite avoiding bilinear pairing, their design did not withstand malicious KGC attacks. To counter coalition attacks, Xu et al. [24] and Han et al. [25] independently proposed two CLAS-based authentication methods that bypass bilinear pairing. Later, Zhu et al. uncovered vulnerabilities in these methods, specifically against malicious KGC and Public Key Replacement (PKR) attacks, and proposed a pairing-free CLAS protocol with anonymity. Their approach allowed only valid single signatures to be aggregated into an aggregate signature, whose legitimacy could be publicly verified [26]. Subsequently, Yang et al. demonstrated that the CLAS scheme [26] was vulnerable to coalition attacks launched by malicious RSUs and vehicles and also susceptible to PKR attacks [27]. In response, they enhanced the security of the CLAS protocol [26] by developing a more efficient CLAS-based authentication scheme. However, their improved protocol still allowed forging of valid aggregate signatures using public information [28]. Consequently, they introduced an anonymous, pairing-free CLAS protocol that is provably secure under random oracles and supports public verification of the aggregate signature's validity.

Research Gaps: Current studies mainly focus on message authentication in VANETs, while overlooking the complete vehicular trust lifecycle. Most schemes do not verify whether the actual driver is a legitimate user before vehicle access, lack password update and recovery mechanisms, provide limited resistance to physical key extraction, and impose considerable computation overhead on resource-constrained OBUs.

III. PRELIMINARIES

This section delves into elliptic curve cryptosystem, physically unclonable function, fuzzy extractors, and secret sharing. Let $\mathbb{N}$ denote the security parameter space and $\lambda \in \mathbb{N}$ the concrete security parameter.

A. Elliptic Curve Cryptosystem (ECC)

Let $E/\mathbb{F}_p$ be an elliptic curve over a prime field $\mathbb{F}_p$ defined by $E: y^2 \equiv x^3 + ax + b \pmod{p}$, where $a, b \in \mathbb{F}_p$ satisfy $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$. The rational points on E along with the point-at-infinity $\mathcal{O}$ form a cyclic group $G = \langle P \rangle$ of prime order q .

- *Point Addition:* For $P = (x_1, y_1), Q = (x_2, y_2) \in G$:

$$P + Q = \begin{cases} \mathcal{O} & \text{if } P = -Q \\ (x_3, y_3) & \text{otherwise} \end{cases}$$

where $x_3 = \lambda^2 - x_1 - x_2, y_3 = \lambda(x_1 - x_3) - y_1$, with slope $\lambda = \frac{y_2 - y_1}{x_2 - x_1}$ for $P \neq \pm Q$, or $\lambda = \frac{3x_1^2 + a}{2y_1}$ for $P = Q$.

- *Scalar Multiplication:* For $k \in \mathbb{Z}_q^*$: $k \cdot P = \underbrace{P + P + \dots + P}_{k \text{ times}}$.

Discrete Logarithm Problem (DLP): For any Probabilistic Polynomial-Time (PPT) adversary $\mathcal{A}$: $\Pr[\mathcal{A}(P, Q = xP) = x] \leq \text{negl}(\lambda), x \xleftarrow{\$} \mathbb{Z}_q^*$.

Computational Diffie-Hellman Problem (CDHP): Given $P, aP, bP \in G$ for random $a, b \in \mathbb{Z}_q^*$, it is computationally infeasible for any PPT adversary $\mathcal{A}$ to compute abP : $\Pr[\mathcal{A}(P, aP, bP) = abP] \leq \text{negl}(\lambda)$.

Parameter Selection: In VLCADA, we adopt a prime-order 256-bit elliptic curve to achieve an approximate 128-bit classical security level, which is widely regarded as an appropriate balance between security strength and computational efficiency for long-term embedded applications. This security level is commonly associated with standardized curves such as NIST P-256 and Curve25519.

B. Physically Unclonable Function

A PUF is a hardware primitive $\Pi = (\text{Gen}, \text{Eval})$ that exploits inherent and uncontrollable manufacturing variations of integrated circuits to generate device-unique responses. It can be represented as: $R = \text{PUF}(C)$ or $\text{PUF} : \{0, 1\}^x \rightarrow \{0, 1\}^y$, where C denotes the challenge and R denotes the corresponding response. For a given device, the same challenge ideally produces a consistent response, while distinct devices generate statistically different responses to the same challenge. In practice, however, PUF outputs may be affected by environmental factors such as temperature variation, supply-voltage fluctuation, electromagnetic interference, aging, and measurement noise. Therefore, practical PUF systems do not assume perfectly identical responses across all operating conditions. Instead, we adopt the standard bounded-noise assumption: for repeated evaluations of the same challenge, the reproduced response differs from the enrolled reference by at most a small Hamming distance δ with overwhelming probability. Formally, $\Pr[\text{HD}(R, R') \leq \delta] \geq 1 - \text{negl}(\lambda)$, where R is the enrolled response and R' is the reproduced response under normal operating conditions. To guarantee stable authentication in VLCADA, fuzzy extractors are employed to correct such bounded errors and reconstruct identical secrets from noisy responses. During system enrollment, only challenge-response pairs satisfying predefined reliability thresholds are selected and stored. Some properties are summarized as follows:

- **Unclonability:** For any PPT adversary $\mathcal{A}$, $\Pr[\text{Eval}_{\mathcal{A}}(C) = \text{Eval}(C)] \leq \text{negl}(\lambda)$.
- **Tamper-Evidence:** Physical probing or invasive modification changes internal characteristics and may distort responses: $\exists \epsilon > 0$, $\text{HD}(\text{Eval}_{\text{pre}}, \text{Eval}_{\text{post}}) \geq \epsilon$.
- **Reliability:** Under admissible environmental conditions, repeated responses remain sufficiently close to the enrolled reference and can be corrected through the fuzzy extractor.

C. Fuzzy Extractor

Fuzzy extractor $\Gamma = (\text{Gen}, \text{Rep})$ is a cryptographic technique for securely handling biometric data [41]. It consists of two main functions:

- $\text{Gen}(\text{Bio}) \rightarrow (\text{secre}, \text{hd})$: Given biometric data Bio , it outputs a secret key $\text{secre} \in \{0, 1\}^\ell$ and an auxiliary parameter $\text{hd} \in \{0, 1\}^*$.
- $\text{Rep}(\text{Bio}', \text{hd}) \rightarrow \text{secre}$: Given biometric data Bio' and auxiliary parameter hd , it outputs the secret key secre if

$\text{HD}(\text{Bio}, \text{Bio}') \leq t$, where t is the predefined maximum error correction capacity.

Correctness: For Bio' such that $\text{HD}(\text{Bio}, \text{Bio}') \leq t$: $\Pr[\text{Rep}(\text{Bio}', \text{hd}) = \text{secre}] \geq 1 - \text{negl}(\lambda)$.

Security: For any distribution W over $\mathcal{M}$ with min-entropy m , secre is pseudorandom: $\text{SD}((\text{secre}, \text{hd}), (U_\ell, \text{hd})) \leq \text{negl}(\lambda)$.

Construction: Let $\mathcal{C}_{\text{BCH}}$ be a BCH code with parameters $[n, k, 2t + 1]$. For PUF response $w \in \{0, 1\}^n$:

- **Generation:** $\text{secre} \leftarrow \text{Ext}(w)$, $\text{hd} \leftarrow \text{Syn}(w) \in \{0, 1\}^{n-k}$, where Ext is a strong randomness extractor.
- **Reproduction:** $\hat{w} \leftarrow \text{Dec}_{\text{BCH}}(w' \oplus \text{Syn}^{-1}(\text{hd}))$, $\text{secre} \leftarrow \text{Ext}(\hat{w})$.

D. Secret Sharing

Secret sharing is a cryptographic technique that divides a secret into multiple shares, distributed among participants. The secret can be reconstructed only when a sufficient number of shares are combined. This ensures that no single participant can access the secret alone, enhancing security. A (t, n) -threshold scheme over $\mathbb{Z}_q$ satisfies:

Correctness: Any $t + 1$ shares reconstruct $s \in \mathbb{Z}_q$: $\exists$ algorithm Rec s.t. $\text{Rec}(\{s_i\}_{i \in S}) = s \forall S \subseteq [n], |S| > t$.

Secrecy: For any $S \subseteq [n]$ with $|S| \leq t$: $\text{SD}(\{s_i\}_{i \in S}, \{U(\mathbb{Z}_q)\}_{i \in S}) \leq \text{negl}(\lambda)$.

Shamir's Scheme: This scheme implements the (t, n) -threshold secret sharing via polynomial interpolation: $f(x) = s + a_1x + \dots + a_tx^t \in \mathbb{Z}_q[x]$, $s_i = f(i)$. The secret s is the constant term of the polynomial $f(x)$, and each share s_i is evaluated at distinct points $i \in [n]$. The secret can be reconstructed using Lagrange interpolation from any $t + 1$ shares.

Reconstruction Process: To reconstruct the secret s , any $t + 1$ participants combine their shares. Using Lagrange interpolation, the polynomial $f(x)$ is reconstructed, and the secret s is obtained by evaluating $f(0)$:

$$s = f(0) = \sum_{i=1}^{t+1} s_i \cdot \prod_{\substack{1 \leq j \leq t+1 \\ j \neq i}} \frac{-x_j}{x_i - x_j}.$$

This process ensures that the secret can only be recovered with the cooperation of at least $t + 1$ participants.

E. Notation Table

For enhanced clarity, Table II provides a comprehensive list of the symbols and notations used throughout this paper, accompanied by their detailed definitions.

IV. SYSTEM MODEL

This section introduces the system architecture, adversary model, and design requirements for our vehicle-side lightweight conditional anonymous double authentication framework.

A. System Architecture

Our proposed framework is built upon a multi-entity architecture, as depicted in Fig. 1, involving four key components: the TA, the Traffic Management Center (TMC), RSUs, and vehicles.

TABLE II
NOTATIONS

Symbol	Description
VANETs	Vehicular Ad Hoc Networks
VLCADA	Vehicle-Side Lightweight Conditional Anonymous Double Authentication
SLCAS	Signer-side Lightweight Conditional Anonymous Signature
DPM	Dynamic Password Mechanism
CPPA	Conditional Privacy-Preserving Authentication
CLC	CertificateLess Cryptosystem
PKI	Public Key Infrastructure
PUF	Physical Unclonable Function
CRP	Challenge-Response Pair
TMC	Traffic Management Center
TA	Trusted Authority
RSU	Road Side Unit
V2I	Vehicle-to-Infrastructure
V2R	Vehicle-to-RSU
V2V	Vehicle-to-Vehicle
(P_{pub}, s)	Primary public/secret key
(ID_i, pwd_i)	User identity/password
$(Vid_i, PVID_i)$	Vehicle identity/pseudonym
$H_0, H_1, H_2, H_3, H_4, H_5, H_6$	Hash function
$\rho_i = (b_i, X_i, t_i)$	Anonymous signature from the vehicle with Vid_i
m_i	Traffic information broadcast by the vehicle with Vid_i
$Report_j^i$	Signed misbehavior report from RSU j on vehicle i
σ_j	RSU j 's digital signature on the report
Rev_i	Revocation certificate issued by TA for vehicle i
CRL	Certificate Revocation List maintained and signed by TA

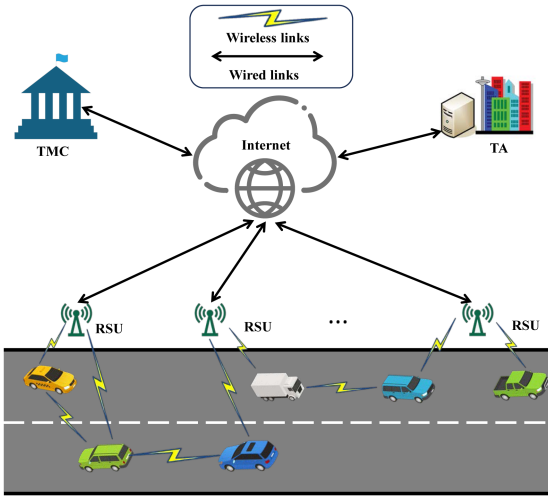


Fig. 1. System Architecture.

TA: The TA serves as the central authority in VANETs, responsible for initializing and managing the system. During the setup phase, the TA generates and distributes system parameters to all vehicles and RSUs. The TA also holds the unique capability to trace the real identity of message senders, ensuring accountability within the network.

TMC: The TMC is a reliable government entity tasked with overseeing urban traffic management. It analyzes real-time traffic data to forecast road conditions and optimize traffic flow, thereby maximizing the utilization of road resources. The TMC also has the authority to instruct the TA to trace the identities of malicious vehicles.

RSUs: RSUs are strategically placed along roadways and intersections as fixed communication infrastructure. They communicate wirelessly with vehicles' OBUs using the DSRC protocol. RSUs can verify the authenticity and integrity of received messages. They can process messages or forward them to the TMC for further analysis and decision-making.

Vehicle: Each vehicle is equipped with an OBU and a PUF to enable secure communication. Vehicles can interact with other vehicles and RSUs through their OBUs, sending and receiving messages wirelessly. The PUF ensures that sensitive information stored on the vehicle cannot be physically stolen.

B. Adversary Model

We adopt a hybrid adversary model that incorporates capabilities from both the Dolev-Yao communication model and the Canetti-Krawczyk security framework, while also extending it to capture emerging real-world threats specific to vehicular systems. The adversary, denoted as $\mathcal{A}$, is characterized by the following capabilities:

- **Communication Control (Dolev-Yao):** $\mathcal{A}$ can fully control insecure V2V and V2I communication channels. This includes eavesdropping, modifying, delaying, replaying, and injecting arbitrary messages between vehicles and RSUs [30].
- **Partial Physical Access (Realistic Adversary Extension):** Vehicles are not assumed to be tamper-proof. $\mathcal{A}$ may gain physical access to a vehicle (e.g., through theft or compromise of a parked vehicle) and attempt to extract sensitive material from its OBU using invasive techniques such as side-channel analysis, fault injection, or memory scraping [8]. Similar adversary assumptions have been adopted in prior work on vehicular tamper resistance, and studies on key extraction have demonstrated the plausibility of such attacks [42].
- **Compromised Authentication Factors (Canetti-Krawczyk + Multi-Factor Threats):** $\mathcal{A}$ may obtain up to $n - 1$ of the authentication factors in a multi-factor authentication system, such as a user's password, biometric template, or PUF-generated token, either through phishing, leakage, or partial sensor spoofing [43]. This assumption aligns with common models in multi-factor security analysis, and our protocol is specifically designed to remain secure unless all n factors are compromised.
- **TA Assumption:** The TA is assumed to be secure and immune to compromise. This is a standard assumption in VANETs and authentication methods [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], as the TA operates in a high-assurance environment.

Algorithm 1: Setup(1^λ).

- 1: **Input:** Security parameter λ
- 2: Select an elliptic curve group G with generator P and order q .
- 3: Randomly choose a primary secret key $s \in \mathbb{Z}_q^*$.
- 4: Compute the primary public key $P_{pub} = s \cdot P$.
- 5: Define four one-way hash functions:
- 6: $H_0 : \{0, 1\}^* \rightarrow \{0, 1\}^{|ID|}$
- 7: $H_i : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$ for $i = 1, 2, 3$.
- 8: **Output:** Primary secret key $msk = s$ and primary public key $mpk = (P, P_{pub}, H_0, H_1, H_2, H_3)$.

C. Design Requirements

VLCADA ought to achieve the following objectives:

- 1) *Authentication and integrity:* The receivers (i.e., vehicles and RSUs) should be able to verify that the received messages are unforged or unmodified and that the sender of the message is a registered and legitimate entity.
- 2) *Conditional anonymity:* VLCADA needs to maintain the confidentiality of vehicle identities to other vehicles and RSUs, while allowing the TA to trace the real identity of vehicles when necessary.
- 3) *Multi-factor security:* VLCADA needs to ensure that the system remains secure even if an attacker gains access to $n - 1$ out of n authentication factors.
- 4) *Resistance to physical attacks:* VLCADA must incorporate measures to safeguard sensitive data from physical attacks, including but not limited to side-channel attacks.
- 5) *Resistance to guessing attacks:* VLCADA is meticulously crafted to protect passwords and derived values against guessing attacks, making it extremely difficult for adversaries to guess these secrets, even with access to stored parameters.
- 6) *Resistance to replay attacks:* VLCADA must implement mechanisms that prevent adversaries from maliciously reusing intercepted transmissions.

V. CONCRETE CONSTRUCTION

This section details the specific development of VLCADA, which comprises two schemes: a SLCAS and a DPM.

A. Signer-Side Lightweight Conditional Anonymous Signature

This subsection presents a signer-side lightweight conditional anonymous signature scheme, primarily used for V2R authentication while ensuring lightweight operations on the vehicle side. The detailed algorithmic construction (Algorithms 1–5) of this scheme is outlined below.

• *Correctness proof:* To prove the correctness of the anonymous signature scheme, we need to verify that the verification equation holds true. Specifically, we need to show that $X = X'$.

1) *Signature equation:*

$$\begin{aligned} b \cdot P &= h_3 \cdot (a + h_2 \cdot sk) \cdot P \\ &= h_3 \cdot (x \cdot Auth^{-1} + h_2 \cdot (x + h_1 \cdot s)) \cdot P \end{aligned}$$

Algorithm 2: PreSign(mpk, s, ID).

- 1: **Input:** Primary public key mpk , primary secret key s , vehicle's real identity ID
- 2: Randomly select $x \in \mathbb{Z}_q^*$.
- 3: Compute:
- 4: $X = x \cdot P$
- 5: $PID = ID \oplus H_0(x \cdot P_{pub})$
- 6: $Auth = H_1(X)$
- 7: $a = x \cdot Auth^{-1}$
- 8: $sk = x + h_1 \cdot s$, where $h_1 = H_1(X, PID, P_{pub})$
- 9: **Output:** $(X, PID, Auth, a, sk)$.

Algorithm 3: Sign(mpk, PID, X, a, sk, m).

- 1: **Input:** Primary public key mpk , PID , X , a , sk , message m
- 2: Generate a timestamp t .
- 3: Compute:
- 4: $h_2 = H_2(PID, X, m, t, 2)$
- 5: $h_3 = H_3(PID, X, m, t, 3)$
- 6: $b = h_3 \cdot (a + h_2 \cdot sk)$
- 7: **Output:** Anonymous signature $\rho = (b, X, t)$.

Algorithm 4: Verify($mpk, PID, m, Auth, \rho$).

- 1: **Input:** Primary public key mpk , PID , message m , $Auth$, signature $\rho = (b, X, t)$
- 2: Verify the timestamp t in ρ .
- 3: **if** timestamp is valid **then**
- 4: Compute:
- 5: $h'_1 = H_1(X, PID, P_{pub})$
- 6: $h'_2 = H_2(PID, X, m, t, 2)$
- 7: $h'_3 = H_3(PID, X, m, t, 3)$
- 8: $X' = Auth \cdot (h'_3 + Auth \cdot h'_2 \cdot h'_3)^{-1} \cdot (b \cdot P - h'_1 \cdot h'_2 \cdot h'_3 \cdot P_{pub})$
- 9: Check if $Auth = H_1(X')$.
- 10: **if** check passes **then**
- 11: **Output:** 1 (valid signature)
- 12: **else**
- 13: **Output:** 0 (invalid signature)
- 14: **end if**
- 15: **else**
- 16: **Output:** 0 (invalid timestamp)
- 17: **end if**

$$\begin{aligned} &= h_3 \cdot (x \cdot Auth^{-1} + h_2 \cdot x + h_2 \cdot h_1 \cdot s) \cdot P \\ &= h_3 x \cdot Auth^{-1} P + h_3 h_2 x \cdot P + h_3 h_2 h_1 \cdot s \cdot P \\ &= Auth^{-1} \cdot h_3 \cdot X + h_2 h_3 \cdot X + h_1 h_2 h_3 \cdot P_{pub} \end{aligned}$$

2) *Verification equation:*

$$\begin{aligned} Auth \cdot bP &= h_3 \cdot X + Auth \cdot h_2 h_3 \cdot X \\ &\quad + Auth \cdot h_1 h_2 h_3 \cdot P_{pub} \end{aligned}$$

Algorithm 5: Trace(mpk, s, ρ, PID).

- 1: **Input:** Primary public key mpk , primary secret key s , X, PID
- 2: Compute:
- 3: $ID = PID \oplus H_0(s \cdot X)$
- 4: **Output:** Real identity ID .

Algorithm 6: PwdSet($ID, pwd, Bio, \{que_j, ans_j\}_{j=1,\dots,n}$).

- 1: **Input:** User name ID , password pwd , biometrics Bio , n question-answer pairs $\{que_j, ans_j\}_{j=1,\dots,n}$
- 2: **Process:**
- 3: Construct a random polynomial $f(x) = k_0 + k_1x + k_2x^2 + \dots + k_{n-1}x^{n-1} \bmod p$, where $k_j \in [0, p-1]$ for $j = 0, 1, \dots, n-1$.
- 4: Generate two one-way hash functions: $H_5 : \{0, 1\}^* \rightarrow \{0, 1\}^{|pwd|}$ and $H_6 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$.
- 5: Compute $(bio, Aux) = Gen(Bio)$.
- 6: Calculate $Tag = H_6(ID, bio, pwd)$ and $c_{pwd} = pwd \oplus H_5(k_0, ID, bio)$.
- 7: For each question-answer pair $\{que_j, ans_j\}_{j=1,\dots,n}$, select random numbers $\{x_j\}_{j=1,\dots,n}$ and compute $\{y_j = f(x_j), Y_j = y_j \oplus H_6(ans_j, ID, bio)\}_{j=1,\dots,n}$.
- 8: **Output:** $(c_{pwd}, Tag, \{que_j, Y_j, x_j\}_{j=1,\dots,n}, H_5, H_6)$.

Algorithm 7: PwdRe($ID, Bio, \{ans_j\}_{j=1,\dots,n}, \{Y_j, x_j\}_{j=1,\dots,n}, c_{pwd}, Tag, \{que_j^{new}, ans_j^{new}\}_{j=1,\dots,n}$).

- 1: **Input:** User name ID , biometrics Bio , answers $\{ans_j\}_{j=1,\dots,n}$, new question-answer pairs $\{que_j^{new}, ans_j^{new}\}_{j=1,\dots,n}$
- 2: **Process:**
- 3: Compute $(bio) = Rep(Bio, Aux)$.
- 4: Calculate $\{y_j = Y_j \oplus H_6(ans_j, ID, bio)\}_{j=1,\dots,n}$.
- 5: Reconstruct the polynomial $f(x) = \sum_{j=1}^n y_j \prod_{1 \leq l \leq n, l \neq j} \frac{(x-x_l)}{(x_j-x_l)} \bmod p$.
- 6: Compute $k_0 = f(0)$ and $pwd = c_{pwd} \oplus H_5(k_0, ID, bio)$.
- 7: Verify $Tag \stackrel{?}{=} H_6(ID, bio, pwd)$. If yes, the password pwd is successfully recovered.
- 8: For each new question-answer pair $\{que_j^{new}, ans_j^{new}\}_{j=1,\dots,n}$, select random numbers $\{x_j^{new}\}_{j=1,\dots,n}$ and compute $\{y_j^{new} = f(x_j^{new}), Y_j^{new} = y_j^{new} \oplus H_6(ans_j^{new}, ID, bio)\}_{j=1,\dots,n}$.
- 9: **Output:** $(pwd, \{Y_j^{new}, x_j^{new}\}_{j=1,\dots,n})$.

3) Reconstructing X :

$$Auth \cdot (b \cdot P - h_1 h_2 h_3 \cdot P_{pub}) = (h_3 + Auth \cdot h_2 h_3) X$$

4) Final reconstruction:

$$X = Auth(h_3 + Auth h_2 h_3)^{-1} (bP - h_1 h_2 h_3 P_{pub}) = X'$$

Thus, the verification equation $Auth = H_1(X')$ holds true, proving the correctness of the anonymous signature scheme.

Algorithm 8: PwdUpd($ID, Bio, c_{pwd}^{old}, pwd^{old}, pwd^{new}, Tag^{old}$).

- 1: **Input:** User name ID , biometrics Bio , old password pwd^{old} , new password pwd^{new} , c_{pwd}^{old} , and Tag^{old}
- 2: **Process:**
- 3: Compute $(bio) = Rep(Bio, Aux)$.
- 4: Verify $Ver^{old} \stackrel{?}{=} H_6(ID, bio, pwd^{old})$. If it does not match, deny the password update.
- 5: Otherwise, compute $c_{pwd}^{new} = c_{pwd}^{old} \oplus pwd^{old} \oplus pwd^{new}$ and $Tag^{new} = H_6(ID, bio, pwd^{new})$.
- 6: **Output:** $(c_{pwd}^{new}, Tag^{new})$.

B. Dynamic Password Mechanism

In this subsection, we present a dynamic password mechanism, which is mainly used to achieve multi-factor access authentication when users access vehicles, as well as the recovery and update of user passwords, to prevent illegal users from accessing vehicle control. The detailed algorithmic construction (Algorithms 6-8) of this scheme is outlined below.

C. Conditional Anonymous Double Authentication Framework

This subsection introduces VLCADA, which integrates the proposed SLCAS and DPM to address the multifaceted security and privacy challenges in VANETs. VLCADA comprises the following sequential stages:

1) *Initialization Phase:* In this phase, the TA initializes the system, including group parameters, primary keys, and hash functions. The steps are as follows:

- ① *System Setup:* The TA executes the setup algorithm $Setup(1^\lambda)$, where λ is the security parameter (e.g., $\lambda = 128$ bits). It selects an additive cyclic group G of prime order q (e.g., $|q| = 256$ bits), with a generator $P \in G$. The TA then samples a primary secret key $s \in \mathbb{Z}_q^*$ uniformly at random and computes the primary public key $P_{pub} = s \cdot P$.
- ② *Hash Function Initialization:* The TA defines a suite of domain-separated cryptographic hash functions, each instantiated using SHA3-256 or HMAC-SHA3 with explicit domain-binding to prevent cross-protocol collisions. These functions serve different roles within the authentication and privacy-preserving operations of the protocol:
 - $H_0 : \{0, 1\}^* \rightarrow \{0, 1\}^{\ell_{ID}}$
Purpose: Generates pseudonymous identifiers for users.
Output Length: $\ell_{ID} = 256$ bits.
Instantiation: SHA3-256.
 - $H_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$
Purpose: Derives secret scalar values for signatures and authentication.
Output Length: Reduced modulo q to match group order.
Instantiation: SHA3-256 with modular reduction.
 - $H_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$
Purpose: Computes challenge values used in anonymous signatures.

Output Length: 256-bit scalar reduced modulo q .
Instantiation: SHA3-256.

– $H_3 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$

Purpose: Generates auxiliary randomness for signature components.

Output Length: 256-bit scalar reduced modulo q .
Instantiation: SHA3-256.

– $H_4 : \{0, 1\}^* \rightarrow \{0, 1\}^{2|q|}$

Purpose: Supports strong key masking and secure password/identity recovery mechanisms.

Output Length: 512 bits, sufficient to mask two concatenated 256-bit values.

Instantiation: SHA3-512.

- ③ *Public Parameter Distribution*: The TA publishes the primary public key and hash function descriptions as the system's public parameters:

$$mpk = (G, q, P, P_{pub}, H_0, H_1, H_2, H_3, H_4).$$

The primary secret key s remains confidential and is securely retained by the TA.

2) *Vehicle Registration Phase*: In this phase, each vehicle and its associated user collaboratively register with the TA to obtain unique authentication credentials. The process integrates biometric and PUF-based device-specific inputs to ensure strong identity binding. The steps are as follows:

- ① *User Input Initialization*: The user locally selects a unique identity string ID_i , a password pwd_i , and provides biometric data Bio_i (e.g., fingerprint or iris scan). These elements ID_i, pwd_i, Bio_i are input into the vehicle's on-board authentication system.
- ② *Vehicle Credential Generation*: Upon receiving the user input, the vehicle performs the following cryptographic operations:
 - Derive a stable biometric key via fuzzy extractor: $(bio_i, Aux_{i,1}) = Gen(Bio_i)$.
 - Generate a physical challenge-response pair (Cha_i, Res_i) using the embedded PUF, where $Res_i = PUF(Cha_i)$.
 - Derive a PUF key via fuzzy extractor: $(res_i, Aux_{i,2}) = Gen(Res_i)$.
 - Compute the vehicle identity and masked password: $Vid_i = H_0(ID_i, bio_i, res_i)$, $Vpwd_i = H_0(pwd_i, bio_i, res_i)$.
 - Send the pair $\{Vid_i, Vpwd_i\}$ to the TA over a secure channel.
- ③ *TA Credential Assignment*: Upon receiving the vehicle's pseudonym credentials, the TA performs the following:
 - Derives a vehicle-specific registration key: $s_i = H_1(Vid_i, Vpwd_i) \oplus s$, where s is the system's primary secret key.
 - Records the tuple $\{Vid_i, Vpwd_i, s_i\}$ in its secure verifier database.
 - Sends s_i back to the vehicle via a secure channel.
- ④ *Vehicle Storage of Authentication Token*:
Upon receiving s_i , the vehicle reconstructs the primary secret key via $s = H_1(Vid_i, Vpwd_i) \oplus s_i$, and generates its local authentication record as follows:

- Compute the local verifier token: $Ver_i = H_1(s, Vid_i, ID_i, pwd_i, bio_i, Aux_{i,1}, res_i, Aux_{i,2})$.
- Mask and bind the internal credentials: $c_i = (s || Vid_i) \oplus H_4(ID_i, pwd_i, bio_i, Aux_{i,1}, res_i, Aux_{i,2})$.
- Securely store the authentication state: $\{Aux_{i,1}, Cha_i, Aux_{i,2}, c_i, Ver_i\}$.

3) *RSU Registration Phase*: In this phase, each RSU registers with the TA to obtain certified cryptographic credentials for verifying vehicular messages and assisting in traceability operations. The procedure is as follows:

- ① *RSU Key Generation and Registration Request*: The RSU generates a long-term key pair by first selecting a unique identity ID_j and a secret key $sk_j \in \mathbb{Z}_q^*$. It then computes the corresponding public key as: $pk_j = sk_j \cdot P$. The RSU submits a registration request to the TA over a secure authenticated channel, consisting of its identity ID_j , public key pk_j , and manufacturer-issued certificate.
- ② *TA Credential Generation and Delivery*: Upon receiving the request, the TA authenticates the RSU based on its manufacturer-issued certificate. If verification succeeds, the TA computes a credential tag for the RSU as: $Cer_j = H_1(s, P_{pub}, ID_j, pk_j)$, where s is the TA's primary secret key. The TA then transmits the signed credential Cer_j to the RSU.
- ③ *RSU Secure Storage Initialization*: After receiving Cer_j , the RSU initializes local secure storage as follows:
 - It generates a PUF challenge-response pair (Cha_j, Res_j) , where $Res_j = PUF(Cha_j)$.
 - Using the biometric helper mechanism, it derives: $(res_j, Aux_j) = Gen(Res_j)$.
 - Computes the verifier tag for local validation: $Ver_j = H_1(ID_j, sk_j, pk_j, Cer_j, Aux_j, res_j)$.
 - Constructs a masked token for secure recovery of credentials: $c_j = (sk_j || Cer_j) \oplus H_4(ID_j, pk_j, Aux_j, res_j)$.
 - Finally, the RSU securely stores the following tuple: $\{Aux_j, Cha_j, c_j, Ver_j\}$.
- 4) *Login Authentication Phase*: In this phase, each vehicle performs a multi-factor authentication procedure to verify the legitimacy of the user attempting to access the vehicle. This process leverages biometric data, password knowledge, and a PUF-assisted storage mechanism. The steps are as follows:
 - ① *User Credential Input*: The user provides their login credentials to the vehicle, including: identity ID'_i , password pwd'_i , and biometric data Bio'_i .
 - ② *Vehicle-Side Verification*: Upon receiving the input, the vehicle proceeds as follows:
 - Reconstructs the biometric template from the auxiliary information: $bio'_i = Rep(Bio'_i, Aux_{i,1})$.
 - Re-generates the PUF response using the stored challenge: $Res'_i = PUF(Cha_i)$.
 - Recovers the reproducible secret from the noisy response: $res'_i = Rep(Res'_i, Aux_{i,2})$.
 - Derives the temporary vehicle secret and pseudonym: $(s' || Vid'_i) = c_i \oplus H_4(ID'_i, pwd'_i, bio'_i, Aux_{i,1}, res'_i, Aux_{i,2})$.

- Computes a new verifier value: $Ver'_i = H_1(s', Vid'_i, ID'_i, pwd'_i, bio'_i, Aux_{i,1}, res'_i, Aux_{i,2})$.
- Compares the computed verifier with the stored value: $Ver'_i \stackrel{?}{=} Ver_i$. If the comparison holds, authentication is successful and access is granted.

5) *Anonymous Authentication Phase*: This phase enables the vehicle to generate a conditional anonymous signature for a message m and broadcast it securely to surrounding RSUs and vehicles. Receiving nodes can independently verify the authenticity and integrity of the message without exposing the sender's real identity. The detailed steps are as follows:

- ① *Offline Signature Preparation*: Prior to actual communication, the vehicle V_i executes the pre-signature algorithm using: $(X_i, PVID_i, Auth_i, a_i, sk_i) = PreSign(mpk, s', Vid'_i)$.
- ② *Signature Generation (Online)*: For each outgoing message m , the vehicle computes the anonymous signature: $\rho_i = (b_i, X_i, t_i) = Sign(mpk, PVID_i, X_i, a_i, sk_i, m)$, where t_i is a current timestamp to prevent replay attacks.
- ③ *Broadcast Message*: The vehicle transmits the package $(m, PVID_i, Auth_i, \rho_i)$ to nearby RSUs and vehicles.
- ④ *Signature Verification*: Upon reception, any verifier (RSU or vehicle) performs $valid \leftarrow Verify(mpk, PVID_i, m, Auth_i, \rho_i)$. If $valid = 1$, the message is accepted as authentic; otherwise, it is discarded.

6) *Malicious Vehicle Tracking and Revocation Phase*: To balance accountability and conditional anonymity, this phase empowers the TA to identify and revoke misbehaving vehicles with the assistance of RSUs. RSUs act as decentralized monitoring agents that detect suspicious behaviors and report them to the TA. The detailed protocol proceeds as follows:

- ① *Misbehavior Detection by RSU*: Each RSU continuously observes the broadcast traffic from nearby vehicles. If abnormal behavior is detected, the RSU constructs a signed report $Report_j^i = \{PVID_i, m, Auth_i, \rho_i, \sigma_j\}$, where $\sigma_j = Sign_{sk_j}(PVID_i || m || Auth_i || \rho_i)$ is the digital signature generated using RSU j 's private key sk_j .
- ② *Report Transmission to TA*: The RSU securely sends the report $Report_j^i$ to the TA over a trusted link. The TA verifies the report by checking the RSU's certificate and validating σ_j .
- ③ *Trace Execution by TA*: Upon successful report, the TA retrieves $\rho_i = (b_i, X_i, t_i)$ from the report and performs identity tracing $VID_i = PVID_i \oplus H_0(s \cdot X_i)$ using its primary secret key s . This tracing mechanism ensures that only the TA can reveal real identities, preserving anonymity for compliant vehicles.
- ④ *Revocation Certificate Generation*: If the behavior is confirmed malicious, the TA issues a digitally signed revocation certificate: $Rev_i = Sign_s(VID_i || reason || timestamp)$, and inserts it into the global Certificate Revocation List (CRL), which is maintained and signed by the TA.
- ⑤ *CRL Dissemination*: The TA periodically broadcasts the latest CRL to the entire VANET, including all RSUs and vehicles. The CRL contains tuples of the form:

$\{VID_i, PVID_i, timestamp\}$, where $PVID_i$ represents previously used pseudonyms. For efficiency, the CRL may be represented using delta updates or Bloom filters to reduce communication and storage overhead.

- ⑥ *Local Enforcement*: Upon receiving the CRL, each RSU and vehicle updates its revocation cache. Any message carrying a pseudonym $PVID_i$ that matches an entry in the CRL is:
 - Immediately discarded;
 - Logged for audit;
 - Prevented from accessing collaborative applications (e.g., routing, alerts, etc.).

This mechanism ensures real-time revocation enforcement while minimizing the performance impact on legitimate participants.

7) *User Password Recovery Phase*: This optional phase enables legitimate users to securely recover their forgotten passwords through multi-factor authentication, incorporating biometrics and knowledge-based questions. The protocol proceeds as follows:

- ① *User Input*: The user interacts with the vehicle interface and provides:
 - Identity string ID ;
 - Biometric sample Bio ;
 - Answers to previously registered security questions: $\{ans_j\}_{j=1,\dots,n}$;
 - A new set of question-answer pairs for future recovery: $\{que_j^{new}, ans_j^{new}\}_{j=1,\dots,n}$.
- ② *Retrieval of Stored Recovery Data*: The vehicle accesses the recovery-related records stored during the *PwdSet* phase, which include: $(\{Y_j, x_j\}_{j=1,\dots,n}, c_{pwd}, Tag)$.
- ③ *Password Reconstruction*: The vehicle executes the *PwdRe* algorithm: $pwd_i, \{Y_j^{new}, x_j^{new}\} \leftarrow PwdRe(ID, Bio, \{ans_j\}, \{Y_j, x_j\}, c_{pwd}, Tag, \{que_j^{new}, ans_j^{new}\})$, and stores new recovery shares $\{Y_j^{new}, x_j^{new}\}$ based on the updated security questions.
- ④ *Secure Storage Update*: The vehicle replaces the old question-answer recovery shares with the newly generated ones, thereby completing the recovery phase and strengthening resilience against future forgotten-password incidents.

8) *User Password Update Phase*: This optional phase enables the user to securely update their password while ensuring the consistency and integrity of the authentication system. The procedure unfolds as follows:

- ① *User Input*: The user enters the current identity string ID_i , the biometric sample Bio_i , the old password pwd_i^{old} , and the new password pwd_i^{new} into the vehicle's interface.
- ② *Retrieval of Stored Credentials*: The vehicle accesses its secure storage to retrieve the user's previously stored password masking data: $(c_{pwd}^{old}, Tag^{old})$.
- ③ *Password Update Execution*: The vehicle locally executes the password update algorithm: $(c_{pwd}^{new}, Tag^{new}) \leftarrow PwdUpd(ID_i, Bio_i, c_{pwd}^{old}, pwd_i^{old}, pwd_i^{new}, Tag^{old})$ and stores the new tuple $(c_{pwd}^{new}, Tag^{new})$ for subsequent login or recovery procedures.

- ④ *Data Replacement*: Upon successful password update, the old password masking data and tag are replaced with their updated counterparts in the vehicle's secure storage.

VI. SECURITY EVALUATION

This section provides a comprehensive security evaluation of the proposed signer-side lightweight conditional anonymous signature scheme. We first establish a formal security model and then present a reductionist proof demonstrating the scheme's resilience against existential forgery under adaptive chosen-message attacks in the Random Oracle Model (ROM). Additionally, we provide an informal security analysis to demonstrate that the scheme satisfies expected security goals and is resistance to several potential attacks.

A. Security Model Definition

The security of the proposed scheme is formalized through a well-defined game between a challenger $\mathcal{C}$ and an adversary $\mathcal{A}$. This game-theoretic approach allows us to precisely quantify the adversary's ability to break the scheme's security properties. The adversary $\mathcal{A}$ is equipped with the power to make several types of queries, each representing a different avenue of attack or information-gathering:

- *Setup-Oracle*: The challenger $\mathcal{C}$ generates the primary secret key and the entire set of system parameters. These parameters form the foundation of the cryptographic scheme, dictating the rules and structures within which the signing and verification processes occur. Once generated, $\mathcal{C}$ sends the system parameters to $\mathcal{A}$, providing the adversary to initiate its attacks.
- *Hash-Oracles* ($H_0 - H_3$): For each of the one-way hash functions H_i ($i = 0, 1, 2, 3$), the challenger $\mathcal{C}$ maintains a list L_{H_i} of tuples (msg, γ) . When the adversary $\mathcal{A}$ makes a query for a message msg to the H_i -oracle, $\mathcal{C}$ first checks if the tuple (msg, γ) already exists in L_{H_i} . If it does, $\mathcal{C}$ returns the precomputed value γ to $\mathcal{A}$. Otherwise, $\mathcal{C}$ selects a random number $\gamma \in \mathbb{Z}_q^*$, inserts the tuple (msg, γ) into L_{H_i} , and then returns γ to $\mathcal{A}$. This mechanism emulates the behavior of a truly random hash function, ensuring that the adversary has no prior knowledge of the hash values and can only obtain them through legitimate queries.
- *Sign-Oracle*: When the adversary $\mathcal{A}$ submits a message m to the Sign-Oracle, the challenger $\mathcal{C}$ generates a valid signature $\rho = (b, X, t)$ according to the signing algorithm of the scheme. This signature is then sent to $\mathcal{A}$, allowing the adversary to analyze the structure of valid signatures.

The adversary $\mathcal{A}$ is considered to have violated the authentication of the scheme if it can generate a valid signature $\rho = (b, X, t)$ without proper authorization. We define $\text{Adv}_S^{\text{Auth}}(\mathcal{A})$ as the probability that $\mathcal{A}$ can achieve this feat.

Definition 1: A conditional anonymous signature scheme S is deemed secure if $\text{Adv}_S^{\text{Auth}}(\mathcal{A})$ is negligible for any polynomial-time adversary $\mathcal{A}$.

This means that as the security parameter λ increases, the probability of the adversary successfully forging a signature becomes vanishingly small, making the scheme practically invulnerable to attacks.

B. Security Proof

We will prove the security of the proposed signer-side lightweight conditional anonymous signature scheme by reduction. Specifically, we will show that if there exists an adversary $\mathcal{A}$ capable of forging a signature, then we can construct a challenger $\mathcal{C}$ that can solve the DL problem with non-negligible probability.

Assume we are given an instance $(P, Q = s \cdot P)$ of the DLP, where P is the generator of an elliptic curve group G of order q , and Q is a point on the curve. The goal of the DLP is to find the integer s such that $Q = s \cdot P$.

The challenger $\mathcal{C}$ simulates the oracles queried by the adversary $\mathcal{A}$ as follows:

- *Setup-Oracle*: $\mathcal{C}$ sets $P_{\text{pub}} \leftarrow Q$ and sends the system parameters $\text{params} = (P, P_{\text{pub}}, H_0, H_1, H_2, H_3)$ to $\mathcal{A}$. By doing so, $\mathcal{C}$ embeds the DL problem instance into the system parameters, creating a connection between the signature-forging problem and the DL problem.
- *H_0 -Oracle*: $\mathcal{C}$ maintains a list L_{H_0} of the form $\langle \delta, \gamma \rangle$, initially empty. Upon receiving a query for message δ , $\mathcal{C}$ checks if $\langle \delta, \gamma \rangle$ exists in L_{H_0} . If it does, $\mathcal{C}$ returns γ to $\mathcal{A}$; otherwise, $\mathcal{C}$ generates a random number $\gamma \in \mathbb{Z}_q^*$, adds $\langle \delta, \gamma \rangle$ to L_{H_0} , and returns γ to $\mathcal{A}$.
- *H_1 -Oracle*: $\mathcal{C}$ maintains a list L_{H_1} of the form $\langle \text{PID}_i, X_i, P_{\text{pub}}, \gamma \rangle$, initially empty. Upon receiving a query for message $\langle \text{PID}_i, X_i, P_{\text{pub}} \rangle$, $\mathcal{C}$ checks if $\langle \text{PID}_i, X_i, P_{\text{pub}}, \gamma \rangle$ exists in L_{H_1} . If it does, $\mathcal{C}$ returns γ to $\mathcal{A}$; otherwise, $\mathcal{C}$ generates a random number $\gamma \in \mathbb{Z}_q^*$, adds $\langle \text{PID}_i, X_i, P_{\text{pub}}, \gamma \rangle$ to L_{H_1} , and returns γ to $\mathcal{A}$.
- *H_2 -Oracle*: $\mathcal{C}$ maintains a list L_{H_2} of the form $\langle \text{PID}_i, X_i, m_i, t_i, \gamma \rangle$, initially empty. Upon receiving a query for message $\langle \text{PID}_i, X_i, m_i, t_i \rangle$, $\mathcal{C}$ checks if $\langle \text{PID}_i, X_i, m_i, t_i, \gamma \rangle$ exists in L_{H_2} . If it does, $\mathcal{C}$ returns γ to $\mathcal{A}$; otherwise, $\mathcal{C}$ generates a random number $\gamma \in \mathbb{Z}_q^*$, adds $\langle \text{PID}_i, X_i, m_i, t_i, \gamma \rangle$ to L_{H_2} , and returns γ to $\mathcal{A}$.
- *H_3 -Oracle*: $\mathcal{C}$ maintains a list L_{H_3} of the form $\langle \text{PID}_i, X_i, m_i, t_i, \gamma \rangle$, initially empty. Upon receiving a query for message $\langle \text{PID}_i, X_i, m_i, t_i \rangle$, $\mathcal{C}$ checks if $\langle \text{PID}_i, X_i, m_i, t_i, \gamma \rangle$ exists in L_{H_3} . If it does, $\mathcal{C}$ returns γ to $\mathcal{A}$; otherwise, $\mathcal{C}$ generates a random number $\gamma \in \mathbb{Z}_q^*$, adds $\langle \text{PID}_i, X_i, m_i, t_i, \gamma \rangle$ to L_{H_3} , and returns γ to $\mathcal{A}$.
- *Sign-Oracle*: When $\mathcal{A}$ submits a message m_i to the Sign-Oracle, $\mathcal{C}$ generates six random numbers $b_i, \epsilon_i, \zeta_i, \eta_i, \delta_i, \text{PID}_i \in \mathbb{Z}_q^*$ and computes $X_i = (b_i \cdot P - \epsilon_i \cdot \zeta_i \cdot \eta_i \cdot P_{\text{pub}})(\delta_i \cdot \eta_i + \zeta_i \cdot \eta_i)^{-1}$. $\mathcal{C}$ adds $\langle \text{PID}_i, X_i, P_{\text{pub}}, \epsilon_i \rangle$ to L_{H_1} , adds $\langle \text{PID}_i, X_i, m_i, t_i, \zeta_i \rangle$ to L_{H_2} , and adds $\langle \text{PID}_i, X_i, m_i, t_i, \eta_i \rangle$ to L_{H_3} . $\mathcal{C}$ then sends $\rho = (b_i, X_i, t_i, m_i)$ to $\mathcal{A}$.

After the simulation, if $\mathcal{A}$ outputs $\rho = (b_i, X_i, t_i, m_i)$, $\mathcal{C}$ checks whether the following equation holds:

$$b_i \cdot P = (\delta_i \cdot \eta_i + \zeta_i \cdot \eta_i) \cdot X_i + \epsilon_i \cdot \zeta_i \cdot \eta_i \cdot P_{\text{pub}}. \quad (1)$$

If the equation does not hold, $\mathcal{C}$ aborts the process. According to the forgery lemma, if $\mathcal{A}$ can output another valid signature $\rho' = (b'_i, X'_i, t'_i, m'_i)$ by repeating the process with a different choice of H_1 , we can obtain the following equation:

$$b'_i \cdot P = (\delta_i \cdot \eta_i + \zeta_i \cdot \eta_i) \cdot X'_i + \epsilon'_i \cdot \zeta_i \cdot \eta_i \cdot P_{\text{pub}}. \quad (2)$$

From equations (1) and (2), we can derive:

$$(b_i - b'_i) \cdot P = (\epsilon_i - \epsilon'_i) \cdot \zeta_i \cdot \eta_i \cdot P_{pub}. \quad (3)$$

Assuming $h_3 \neq h'_3$, we can solve for s :

$$s = (b_i - b'_i)((\epsilon_i - \epsilon'_i) \cdot \zeta_i \cdot \eta_i)^{-1}. \quad (4)$$

This contradicts the hardness of the DLP. Therefore, the proposed signer-side lightweight conditional anonymous signature scheme is secure against forgery under adaptive chosen-message attacks in the random oracle model.

C. Security Analysis

This section provides a security analysis to demonstrate that VLCADA achieves the expected security goals and is resilient to various potential attacks.

Authentication and integrity: The authentication and integrity of VLCADA scheme are guaranteed by the underlying cryptographic primitives and the hardness of the DLP. Specifically, based on the formal security proof presented in Section VI-B, any adversary $\mathcal{A}$ cannot successfully forge or tamper with a signature $\rho = (b, X, t)$ such that it would be verified and accepted by the corresponding receiver. This is because the DLP, which underpins the security of our signature scheme, is computationally infeasible to solve. Consequently, VLCADA achieves both authentication and integrity.

Conditional anonymity: VLCADA ensures conditional anonymity by employing a pseudonym $PID_i = ID_i \oplus H_0(x_i \cdot P_{pub})$ to conceal the vehicle's real identity ID_i . Here, x_i is a random number known only to the vehicle, and $P_{pub} = s \cdot P$. To extract ID_i , an adversary $\mathcal{A}$ would need to compute $x_i \cdot P_{pub} = x_i \cdot s \cdot P$ from $X_i = x_i \cdot P$ and $P_{pub} = s \cdot P$. This requires either obtaining the secret s or solving the CDHP, both of which are assumed to be infeasible. Since only the TA has access to s and the CDHP is hard, the adversary $\mathcal{A}$ cannot deduce the vehicle's real identity from the pseudonym. Moreover, the TA can trace the real identity of a malicious vehicle by computing $ID_i = PID_i \oplus H_0(s \cdot X_i)$ when necessary. This design ensures that the scheme achieves conditional anonymity while allowing for accountability in the presence of malicious behavior.

Multi-factor security: VLCADA employs a multi-factor security mechanism, ensuring that the system remains secure even if an adversary $\mathcal{A}$ compromises $n - 1$ out of n security factors. The multi-factor security encompasses a combination of the password pwd_i , the owner's biometric information Bio_i , and PUF information $(Cha_i, Aux_{i,2})$. The following cases illustrate the robustness of this mechanism:

- *Case 1:* If an adversary $\mathcal{A}$ possesses the owner's biometric information Bio_i and PUF information $(Cha_i, Aux_{i,2})$, it still cannot obtain the composite identifier $(s' || Vid'_i) = c_i \oplus H_4(ID_i, pwd_i, bio'_i, Aux_{i,1}, res'_i, Aux_{i,2})$ due to the lack of the password pwd_i .
- *Case 2:* If an adversary $\mathcal{A}$ has obtained the password pwd_i and PUF information $(Cha_i, Aux_{i,2})$, it still cannot derive the composite identifier $(s' || Vid'_i) = c_i \oplus$

$H_4(ID_i, pwd_i, bio'_i, Aux_{i,1}, res'_i, Aux_{i,2})$ due to the absence of the owner's biometric information Bio_i .

- *Case 3:* Even if an adversary has both the password pwd_i and the owner's biometric information Bio_i , it cannot obtain the composite identifier $(s' || Vid'_i) = c_i \oplus H_4(ID_i, pwd_i, bio'_i, Aux_{i,1}, res'_i, Aux_{i,2})$ without access to the PUF information $(Cha_i, Aux_{i,2})$.

In summary, the multi-factor security design ensures that the remaining factors remain secure even if any two out of the three are compromised.

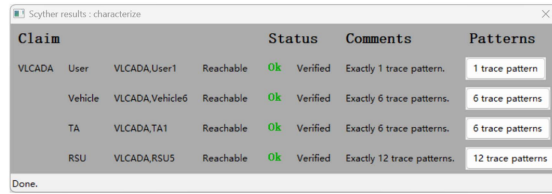
Resilience to cloning and physical attacks: VLCADA is designed to be resilient against cloning and physical attacks. Suppose an adversary $\mathcal{A}$ gains access to the parameters $\{Aux_{i,1}, Cha_i, Aux_{i,2}, c_i, Ver_i\}$ stored on the vehicle through physical attacks. To forge a valid signature, the adversary would need to compute $c_i = (s || Vid_i) \oplus H_4(ID_i, pwd_i, bio_i, Aux_{i,1}, res_i, Aux_{i,2})$, $(bio_i, Aux_{i,1}) = Gen(Bio_i)$, $Res_i = PUF(Cha_i)$, and $(res_i, Aux_{i,2}) = Gen(Res_i)$. However, the adversary cannot calculate the correct secret parameters $Res_i = PUF(Cha_i)$ and $(res_i, Aux_{i,2}) = Gen(Res_i)$ due to the unclonable nature of PUF. Consequently, the adversary cannot generate a valid signature using the compromised parameters. The vehicle can detect such attempts and take appropriate actions. Therefore, VLCADA is resilient against cloning and physical attacks.

Resilience to guess attacks: VLCADA is designed to withstand guessing attacks. Suppose a vehicle is stolen or lost and falls into the wrong hands. An adversary $\mathcal{A}$ may attempt to exploit this situation by retrieving the stored parameters $\{Aux_{i,1}, Cha_i, Aux_{i,2}, c_i, Ver_i\}$. To deduce the password, the adversary would need to compute $Ver_i = H_1(s, Vid_i, ID_i, pwd_i, bio_i, Aux_{i,1}, res_i, Aux_{i,2})$. This computation requires access to the critical information $\{s, bio_i, res_i\}$, which is not stored on the vehicle and is only known to the legitimate user. Without these essential components, the adversary cannot feasibly deduce the password through brute force or guessing attacks. Therefore, VLCADA is resilient against guessing attacks.

Resistance to replay attacks: VLCADA incorporates timestamps in the anonymous signature $\rho_i = (b_i, X_i, t_i)$ to ensure message freshness. This mechanism allows recipients to verify and dismiss outdated or duplicated messages, effectively preventing replay attacks and enhancing the overall security of VLCADA.

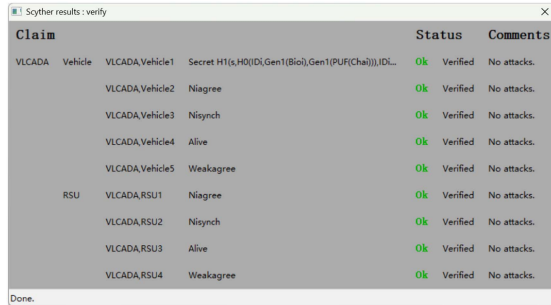
D. Security Verification Using Automated Tool

In this subsection, we present the formal security verification of VLCADA using the *Scyther* tool. *Scyther* is a widely adopted automated formal analysis tool for cryptographic protocols, operating under the assumption of perfect cryptography. This means adversaries are unable to extract any information from encrypted messages without the appropriate decryption keys. By expressing protocols in the Security Protocol Description Language (SPDL), *Scyther* can systematically analyze protocol logic, uncover potential construction flaws, and verify critical security properties. For our analysis, we employed the *scyther-w32-Compromise-0.9.2* version with default



Claim	Status	Comments	Patterns
VLCADA User VLCADA,User1	Reachable Ok Verified	Exactly 1 trace pattern.	1 trace pattern
Vehicle VLCADA,Vehicle6	Reachable Ok Verified	Exactly 6 trace patterns.	6 trace patterns
TA VLCADA,TA1	Reachable Ok Verified	Exactly 6 trace patterns.	6 trace patterns
RSU VLCADA,RSU5	Reachable Ok Verified	Exactly 12 trace patterns.	12 trace patterns

(a) Status



Claim	Status	Comments
VLCADA Vehicle VLCADA,Vehicle1	Secret H1(s,H0ID,Gen1(Bio),Gen1(PUF(Cha))),IDL...	Ok Verified No attacks.
VLCADA,Vehicle2	Niagree	Ok Verified No attacks.
VLCADA,Vehicle3	Nisynch	Ok Verified No attacks.
VLCADA,Vehicle4	Alive	Ok Verified No attacks.
VLCADA,Vehicle5	Weakagree	Ok Verified No attacks.
RSU VLCADA,RSU1	Niagree	Ok Verified No attacks.
VLCADA,RSU2	Nisynch	Ok Verified No attacks.
VLCADA,RSU3	Alive	Ok Verified No attacks.
VLCADA,RSU4	Weakagree	Ok Verified No attacks.

(b) Result

Fig. 2. Verification result.

configurations. As shown in Fig. 2, the Scyther verification results confirm that VLCADA is secure against a wide spectrum of common adversarial capabilities. These results provide strong evidence that the proposed framework achieves its intended security objectives and is formally validated against realistic vehicular threat models.

VII. PERFORMANCE EVALUATION

In this section, we evaluate the performance of VLCADA, focusing on its computational and communication overhead in comparison to a range of representative baseline schemes [18], [25], [26], [27], [28], [29], [31], [32], [33], [34], [35], [36].

A. Computation Overhead

To reflect the heterogeneous nature of VANET deployment environments, where vehicles and RSUs operate under differing constraints, we adopt a dual-hardware simulation platform:

- *Vehicle-side emulator*: A Virtual Machine (VM) running on a host system equipped with an AMD Ryzen 7 8845HS CPU (3.80 GHz, 16 logical cores) and 32 GB of RAM. The VM is configured with 5 logical cores and 12.00 GiB of RAM, running Ubuntu 18.04 LTS. This setup emulates a resource-constrained OBU, capturing the trade-off between computational feasibility and the limitations typically encountered in real-world devices. The configuration setup is illustrated in Fig. 3.
- *RSU-side emulator*: A Raspberry Pi 4 Model B with a quad-core Cortex-A72 CPU (1.80 GHz) and 4 GB RAM is used to simulate a resource-constrained RSU, reflecting realistic hardware profiles for edge-deployed devices in intelligent transportation systems. A physical photo of the Raspberry is shown in Fig. 4.

All cryptographic primitives, such as hash functions, ECC operations, PUF, and fuzzy extractors, were implemented in

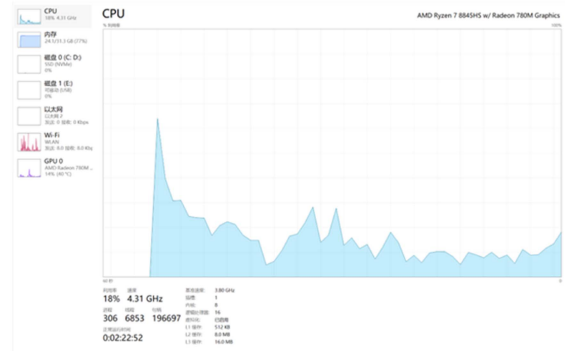


Fig. 3. Notebook configuration.

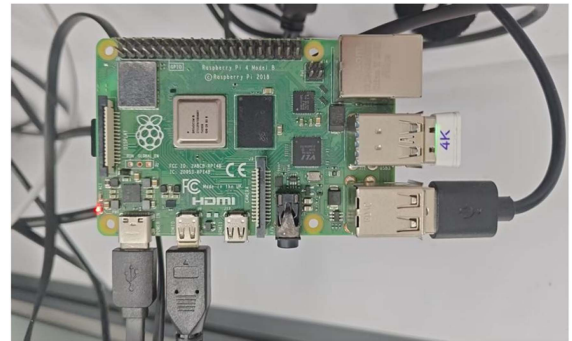


Fig. 4. Raspberry.

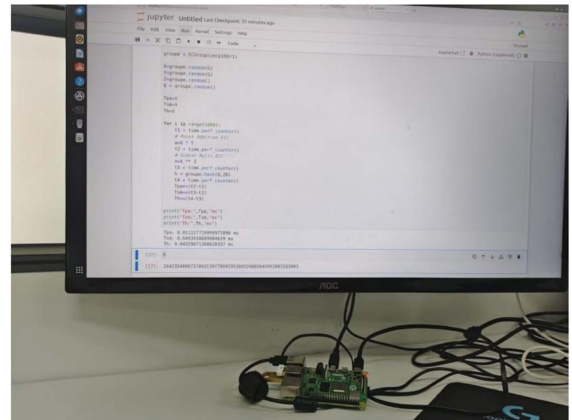


Fig. 5. Experimental physical diagram.

Python using the charm-crypto library [44]. To ensure accuracy and repeatability, we measured the execution time of each basic cryptographic operation independently on both platforms. The experimental physical diagram is shown in Fig. 5. The results are summarized in Table III. We evaluate the VLCADA's computation overhead during its core online operations, signature generation and signature verification, as these are the common functionalities implemented across all baseline schemes. Auxiliary operations such as login authentication and credential recovery, while supported in VLCADA, are excluded from cross-scheme comparisons to ensure fairness. In the VLCADA, the vehicle only performs two hash computations to generate a pseudonymous message signature, resulting in a cost

TABLE III
EXECUTION TIME OF BASIC OPERATIONS

Description	Run Time (ms)	
	Vehicle	RSU
Hash operation	$T_{h-v} = 0.003$	$T_{h-r} = 0.043$
Point addition on ECC	$T_{pa-v} = 0.001$	$T_{pa-r} = 0.011$
Scalar multiplication on ECC	$T_{m-v} = 0.126$	$T_{m-r} = 0.649$
PUF operation	$T_{pu-v} = 0.001$	$T_{pu-r} = 0.005$
Fuzzy generation operation	$T_{fe-v} = 0.41$	$T_{fe-r} = 5.16$
Fuzzy reproduction operation	$T_{re-v} = 0.202$	$T_{re-r} = 2.219$

of: $2T_{h-v} = 0.006$ ms. On the RSU side, the signature verification involves two ECC scalar multiplications, one point addition, and four hash operations: $2T_{sm-r} + T_{pa-r} + 4T_{h-r} = 1.481$ ms. To benchmark the VLCADA against representative schemes [18], [25], [26], [27], [28], [29], [31], [32], [33], [34], [35], [36], we adopt a unified computation model, applying the operation counts reported in each scheme to our platform-specific latency measurements. Table IV summarizes the resulting performance metrics. Obviously, the VLCADA demonstrates superior efficiency over the majority of existing approaches.

B. Communication Overhead

We evaluate the communication overhead of VLCADA and compare it against representative authentication schemes [18], [25], [26], [27], [28], [29], [31], [32], [33], [34], [35], [36]. Let $|Z_q^*|$, $|G|$, $|m|$, $|t|$, and $|ID|$ represent the sizes (in bytes) of a scalar in Z_q^* , a group element in G (generated via ECC), a message m , a timestamp t , and a real identity ID , respectively. For this evaluation, we assume: $|Z_q^*| = |m| = |ID| = 20$ bytes, $|G| = 40$ bytes, $|t| = 4$ bytes. In VLCADA, the vehicle transmits the message $\{m, PVID_i, Auth_i, \rho = (b, X, t)\}$ to the RSU, resulting in a communication overhead of: $|m| + |ID| + 2|Z_q^*| + |G| + |t| = 124$ bytes. Using the same calculation method, the communication overheads of the other baseline schemes [18], [25], [26], [27], [28], [29], [31], [32], [33], [34], [35], [36] are summarized in Table V. Obviously, VLCADA consistently maintains significantly lower communication overhead than all alternative approaches.

C. Network Simulation

To further evaluate our scheme's communication performance in VANETs, we ran simulations using NS3 (version ns-3.27) integrated with the SUMO traffic generator. The simulations were performed on an Ubuntu 18.04 system running on a 3.6 GHz AMD R7-5800H processor, specifically to measure the Packet Delivery Ratio (PDR) of authentication data. Using SUMO, we selected a 3×2.5 km² area near Jiaxing, China (see Fig. 6) and generated a corresponding vehicle mobility trace file. The simulation duration was 318 seconds, involving 57 vehicle nodes. This mobility trace file was then imported into NS3 for the network simulation. Detailed simulation parameters are given in Table VI. PDR refers to the ratio of the total number of data

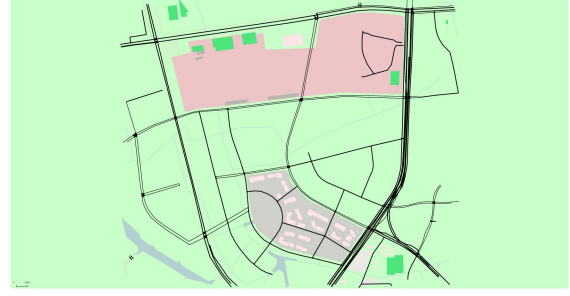


Fig. 6. Simulation scenario.

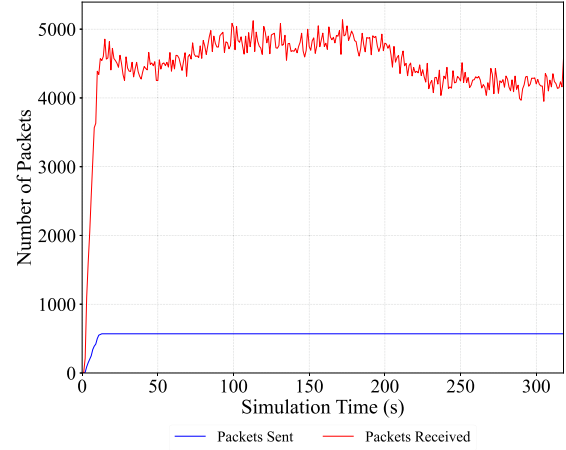


Fig. 7. The variation trend of packets sent and received of VLCADA over the simulation time.

packets successfully received by the destination node to the total number sent by the source node.

Fig. 7 illustrates the variation in the number of data packets sent and received of VLCADA over the simulation time. It can be observed that the number of sent packets increases steadily and gradually stabilizes over time, while the number of received packets rises rapidly from zero during the initial simulation phase and then enters a fluctuating stage due to factors such as communication distance, node mobility, and channel conditions. Fig. 8 illustrates the PDR performance of VLCADA over simulation time, with the results disaggregated into five communication distance intervals to provide a granular analysis of how performance varies with range. In the legend, the data series labeled PDR1 through PDR5 correspond to transmissions within the 0–100 m, 100–200 m, 200–300 m, 300–400 m, and 400–500 m ranges, respectively. As shown, the performance of the VLCADA shows significant degradation with distance. For PDR1, the value stabilizes at ~ 0.82 , while for PDR5, it drops to around 0.64. This drop is a direct result of increasing signal attenuation and path loss. Besides, the initial PDR fluctuations are consistent with the network initialization phase in Fig. 9, where vehicle nodes and communication channels are being established.

Then, we further evaluated the performance of various schemes [18], [25], [26], [27], [28], [29], [31], [32], [33], [34], [35], [36] through the same simulation settings given in Table VI. Besides, all schemes adopted the same stream of

TABLE IV
COMPARISON OF COMPUTATION OVERHEAD

Scheme	Signature Generation	Signature Verification	Total (ms)	Our Improvements Over This Scheme
ID-based method [18]	$2T_{m-v} + 2T_{h-v} = 0.258$	$2T_{m-r} + 2T_{pa-r} + 2T_{h-r} = 1.406$	1.664	10.64%
CLC-based method [25]	$T_{m-v} + T_{h-v} = 0.129$	$2T_{m-r} + T_{pa-r} + 2T_{h-r} = 1.395$	1.524	2.43%
CLC-based method [26]	$T_{m-v} + 2T_{h-v} = 0.132$	$4T_{m-r} + 3T_{pa-r} + 3T_{h-r} = 2.758$	2.89	48.55%
CLC-based method [27]	$T_{m-v} + 2T_{h-v} = 0.132$	$3T_{m-r} + 3T_{pa-r} + 3T_{h-r} = 2.109$	2.241	33.65%
CLC-based method [28]	$T_{m-v} + 2T_{h-v} = 0.132$	$4T_{m-r} + 3T_{pa-r} + 3T_{h-r} = 2.758$	2.89	48.55%
ID-based method [29]	$3T_{m-v} + 3T_{h-v} = 0.387$	$3T_{m-r} + 2T_{pa-r} + 2T_{h-r} = 1.406$	1.793	17.07%
ID-based method [31]	$T_{h-v} = 0.003$	$3T_{m-r} + 2T_{pa-r} + 3T_{h-r} = 2.098$	2.101	29.22%
ID-based method [32]	$T_{m-v} + 3T_{h-v} = 0.135$	$3T_{m-r} + 2T_{pa-r} + 3T_{h-r} = 2.098$	2.233	33.41%
CLC-based method [33]	$T_{h-v} = 0.003$	$3T_{m-r} + 2T_{pa-r} + 2T_{h-r} = 2.055$	2.058	27.75%
ID-based method [34]	$T_{m-v} + T_{h-v} = 0.129$	$3T_{m-r} + 2T_{pa-r} + 2T_{h-r} = 2.055$	2.184	31.91%
CLC-based method [35]	$T_{m-v} + T_{h-v} = 0.129$	$3T_{m-r} + 2T_{pa-r} + T_{h-r} = 2.012$	2.141	30.55%
CLC-based method [36]	$T_{m-v} + T_{h-v} = 0.129$	$3T_{m-r} + 3T_{pa-r} + 2T_{h-r} = 2.066$	2.195	32.26%
Ours	$2T_{h-v} = 0.006$	$2T_m + T_{pa} + 4T_{h-r} = 1.481$	1.487	-

TABLE V
COMPARISON OF COMMUNICATION OVERHEAD

Scheme	Message Content	Message Size (bytes)	Our Improvements Over This Scheme
ID-based method [18]	$\{PID_i, K_i, m_i, \sigma_i = (\sigma_{i,1}, \sigma_{i,2}), T_i\}$	$ m + ID + 2 Z_q^* + 2 G + t = 164$	24.39%
CLC-based method [25]	$\{PID_i, vpk_{PID_i}, m_i, \sigma_i = (S_i, U_i), T_i\}$	$ m + ID + Z_q^* + 2 G + t = 144$	13.89%
CLC-based method [26]	$\{ID_i, vpk_i = (R_i, B_i), m_i, \sigma_i = (\delta_i, U_i), t_i\}$	$ m + ID + Z_q^* + 3 G + t = 184$	32.61%
CLC-based method [27]	$\{ID_i, vpk_i = (R_i, B_i), m_i, \sigma_i = (\delta_i, U_i), t_i\}$	$ m + ID + Z_q^* + 3 G + t = 184$	32.61%
CLC-based method [28]	$\{PID_i, PK_i = (R_i, X_i), m_i, \sigma_i = (\omega_i, U_i), t_i\}$	$ m + ID + Z_q^* + 3 G + t = 184$	32.61%
ID-based method [29]	$\{AID_i = (AID_{i,1}, AID_{i,2}), R_i, M_i, \sigma_i, T_i\}$	$ m + ID + Z_q^* + 2 G + t = 144$	13.89%
ID-based method [31]	$\{AID_i, R_i, m_i, \sigma_i = (d_{j^*}, z), T_i\}$	$ m + ID + 2 Z_q^* + G + t = 124$	00.00%
ID-based method [32]	$\{c'_{j,i}, PID_{j,i}, T_{j,i}, S_{j,i}, \Theta_{j,i}, t_{j,i}, \beta_{j,i}\}$	$ m + ID + 3 Z_q^* + G + t = 144$	13.89%
CLC-based method [33]	$\{PID_i, PK_i, U_i, m_i, t_i, \sigma_i\}$	$ m + ID + Z_q^* + 2 G + t = 144$	13.89%
ID-based method [34]	$\{M_i^j, PID_a^j, AD_j, W_a^j, \sigma_a^j, T_a^j\}$	$ m + ID + 2 Z_q^* + G + t = 124$	00.00%
CLC-based method [35]	$\{PID_i, m_i, pk_i, \sigma_i, T_i\}$	$ m + ID + 2 Z_q^* + 2 G + t = 164$	24.39%
CLC-based method [36]	$\{PID_{i,j}, m, pk_{i,j}, \sigma_{i,j} = (B_{i,j}, z_{i,j}), t_{i,j}\}$	$ m + ID + Z_q^* + 2 G + t = 144$	13.89%
Ours	$\{m, PVID_i, Auth_i, \rho = (b, X, t)\}$	$ m + ID + 2 Z_q^* + G + t = 124$	-

TABLE VI
SIMULATION PARAMETERS

Parameter	Value
Simulation area	$3 \times 2.5 \text{ km}^2$
Simulation time	318 simulated seconds
Vehicle nodes	57
Propagation loss model	The two-ray ground reflection model
Fading model	The Nakagami model
Data transmission rate	6 Mbps
Transmission power	20 dBm
Transmit time interval	100 ms

random numbers, which consequently resulted in consistent PDR data for schemes [31], [34], and VLCADA, given their identical authentication data sizes. For each scheme, we selected the PDR data from the 50th to 300th seconds and averaged them to compare their performance. Although the overall PDR performance of different schemes seems similar under given parameters, the detailed data in Fig. 9 shows that compared to other schemes, VLCADA along with [31], [34] consistently achieve better packet delivery ratios compared to alternative approaches [18], [25], [26], [27], [28], [29], [32], [33], [35], [36].

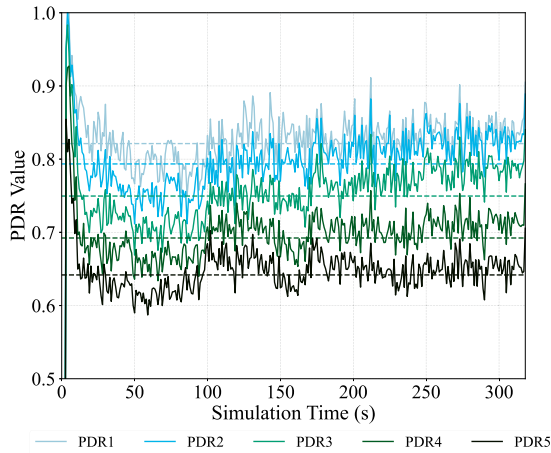


Fig. 8. The variation trend of PDR values of VLCADA over the simulation time under different communication distances.

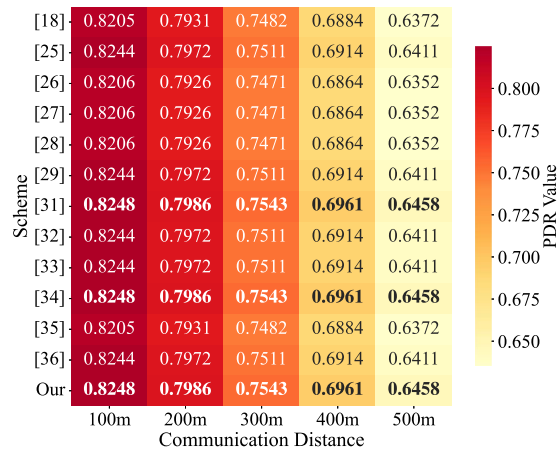


Fig. 9. Comparison of average PDR values among different schemes under various communication distances.

VIII. CONCLUSION

In this paper, we presented VLCADA to address the pressing security and privacy challenges in VANETs. The proposed framework adopts a signer-side lightweight conditional anonymous signature scheme, which shifts computationally intensive operations to an offline pre-computation phase. As a result, the real-time authentication process requires only lightweight operations, significantly reducing the computational burden on resource-constrained vehicles. In addition, VLCADA incorporates a dynamic password mechanism based on secret sharing, enabling secure password update and recovery without requiring intervention from a trusted authority.

Furthermore, VLCADA integrates multi-factor authentication by combining passwords, biometrics, and PUF-based hardware features to provide layered security protection. Even if one authentication factor is compromised, the remaining factors continue to preserve the security of the system. The incorporation of PUF together with fuzzy extractors further strengthens resistance against physical attacks while mitigating the impact of noisy hardware responses.

Comprehensive security analyses and performance evaluations demonstrate that VLCADA satisfies the stringent security and privacy requirements of VANETs while significantly reducing computational and communication overhead compared with existing CPPA schemes. Overall, the VLCADA provides a unified and practical security framework for future intelligent transportation systems. As future work, we plan to investigate a post-quantum extension of VLCADA by integrating standardized quantum-resistant cryptographic primitives, such as ML-KEM and ML-DSA.

ACKNOWLEDGMENT

The authors would like to thank the associate editor and the anonymous reviewers for their valuable comments.

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